



**Topic Test: OxfordAQA**  
**International A level Chemistry**  
AS Organic

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date: \_\_\_\_\_

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Time: **80 minutes**

Marks: **70 marks**

Comments:

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**1**

How many structural isomers have the molecular formula  $C_4H_9Br$ ?

- A**    2    ☐
- B**    3    ☐
- C**    4    ☐
- D**    5    ☐

(Total 1 mark)

**2**

Octane is the eighth member of the alkane homologous series.

(a) State **two** characteristics of a homologous series.

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(2)

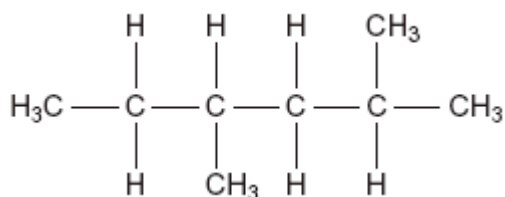
(b) Name a process used to separate octane from a mixture containing several different alkanes.

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(1)

- (c) The structure shown below is one of several structural isomers of octane.



Give the meaning of the term structural isomerism.  
Name this isomer and state its empirical formula.

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(4)

- (d) Suggest why the branched chain isomer shown above has a lower boiling point than octane.

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(2)

(Total 9 marks)

- 3** Which molecule is **not** produced when ethane reacts with bromine in the presence of ultraviolet light?

A  $\text{C}_2\text{H}_4\text{Br}_2$  ☐

B  $\text{HBr}$  ☐

C  $\text{H}_2$  ☐

D  $\text{C}_4\text{H}_{10}$  ☐

(Total 1 mark)

**4**

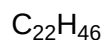
This question is about alkanes with the molecular formulas shown.



**A**



**B**



**C**



**D**

- (a) Give the IUPAC name of an alkane that could be **A**.

\_\_\_\_\_

**(1)**

- (b) Draw skeletal formulas for **two** branched-chain isomers of **B**.

State the type of isomerism shown.

Isomer 1

Isomer 2

Type of isomerism \_\_\_\_\_

**(3)**

- (c) Thermal cracking of one molecule of **C** can form one molecule of **B** and two molecules of another organic compound.

The reaction occurs at a high temperature.

Write an equation for this reaction.

State **one other** reaction condition needed.

Equation

\_\_\_\_\_

Reaction condition \_\_\_\_\_

**(2)**

- (d) Under different conditions, catalytic cracking of **C** can occur as shown.



Name the catalyst used.

\_\_\_\_\_

(1)

**D** reacts with bromine gas in UV light or bright sunlight in the same way as methane reacts with chlorine. The reaction is a free-radical substitution.

- (e) Write an equation for the initiation step of this reaction.

\_\_\_\_\_

(1)

- (f) Write an equation to show the formation of a hydrocarbon in a termination step for the reaction of **D** with bromine gas in UV light.

\_\_\_\_\_

(1)

(Total 9 marks)

5

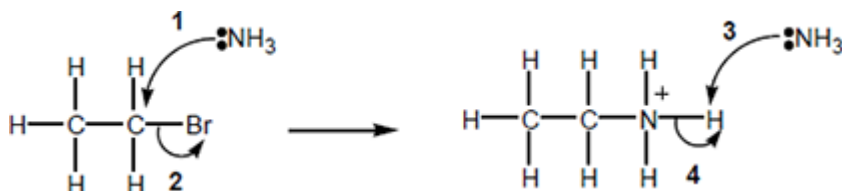
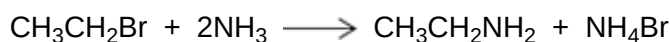
Why are fluoroalkanes unreactive?

- A** Fluorine is highly electronegative. ☐
- B** The  $\text{F}^-$  ion is very stable. ☐
- C** They are polar molecules. ☐
- D** The C–F bond is very strong. ☐

(Total 1 mark)

6

This question is about a method that can be used to prepare ethylamine.



Which statement about the reaction is **not** correct?

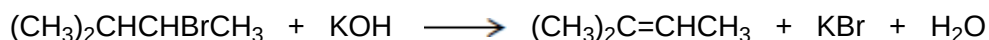
- A** Ethylamine is a primary amine. ☐
- B** The mechanism is a nucleophilic substitution. ☐
- C** Using an excess of bromoethane will prevent further reaction to form a mixture of amine products. ☐
- D** Ammonium bromide is an ionic compound. ☐

(Total 1 mark)

7

Haloalkanes are used in the synthesis of other organic compounds.

- (a) Hot concentrated ethanolic potassium hydroxide reacts with 2-bromo-3-methylbutane to form two alkenes that are structural isomers of each other. The major product is 2-methylbut-2-ene.
- (i) Name and outline a mechanism for the conversion of 2-bromo-3-methylbutane into 2-methylbut-2-ene according to the equation.



Name of mechanism \_\_\_\_\_

Mechanism

(4)

(ii) Draw the **displayed formula** for the other isomer that is formed.

(1)

(iii) State the type of structural isomerism shown by these two alkenes.

\_\_\_\_\_

(1)

- (b) A small amount of another organic compound, **X**, can be detected in the reaction mixture formed when hot concentrated ethanolic potassium hydroxide reacts with 2-bromo-3-methylbutane.

Compound **X** has the molecular formula  $C_5H_{12}O$  and is a secondary alcohol.

(i) Draw the **displayed formula** for **X**.

(1)

(ii) Suggest **one** change to the reaction conditions that would increase the yield of **X**.

\_\_\_\_\_  
\_\_\_\_\_

(1)

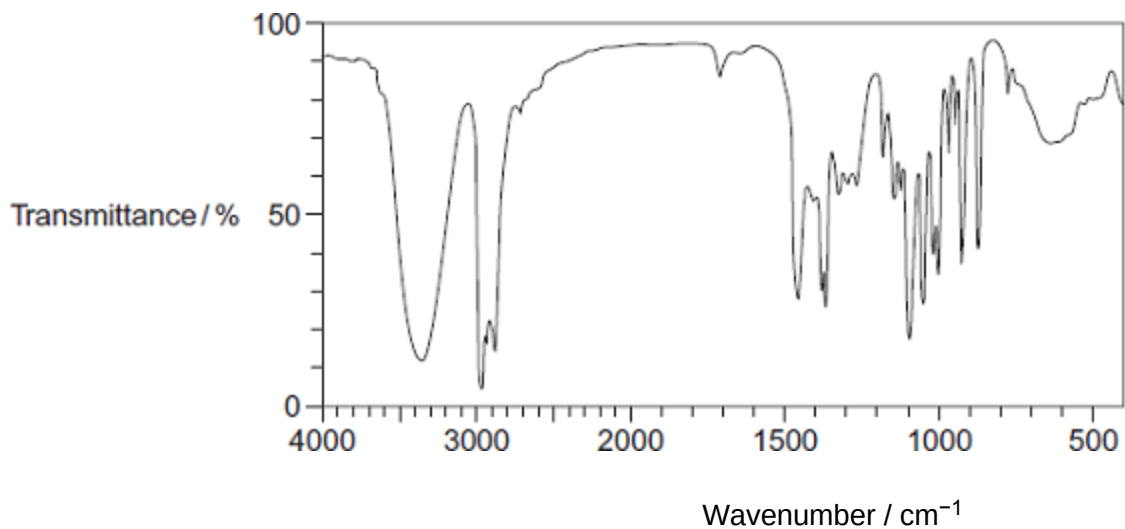
(iii) State the type of mechanism for the conversion of 2-bromo-3-methylbutane into **X**.

\_\_\_\_\_

(1)

- (iv) Identify **one** feature of this infrared spectrum of a pure sample of **X** that may be used to confirm that **X** is an alcohol.

You may find it helpful to refer to **Table 1** on the Data Sheet.



Feature \_\_\_\_\_  
\_\_\_\_\_

(1)

(Total 10 marks)

8

Which statement about ethene is correct?

- A** It has no geometric isomers because there is free rotation around the C=C bond. ☐
- B** It reacts with HBr in a nucleophilic addition reaction. ☐
- C** It burns in excess oxygen to produce carbon dioxide and water. ☐
- D** The C=C bond is twice as strong as the C–C bond in ethane. ☐

(Total 1 mark)

9

Which statement about *E*-1,2-dichloroethene is correct?

- A** It has the same boiling point as *Z*-1,2-dichloroethene. ☐
- B** It forms a polymer with the same repeating unit as *Z*-1,2-dichloroethene. ☐
- C** It has the same IR spectrum as *Z*-1,2-dichloroethene in the range 400–1500 cm<sup>-1</sup>. ☐
- D** It has a molecular ion peak different from that of *Z*-1,2-dichloroethene in its mass spectrum. ☐

(Total 1 mark)



**10**

It is possible to convert but-1-ene into its structural isomer but-2-ene.

- (a) State the type of structural isomerism shown by but-1-ene and but-2-ene.

\_\_\_\_\_

(1)

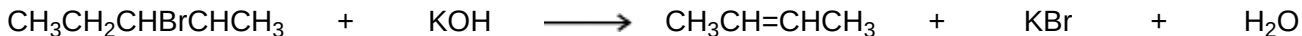
- (b) The first stage in this conversion involves the reaction of hydrogen bromide with but-1-ene.



Outline a mechanism for this reaction.

(4)

- (c) The second stage is to convert 2-bromobutane into but-2-ene.



Outline a mechanism for this reaction.

(3)

(Total 8 marks)

**11**

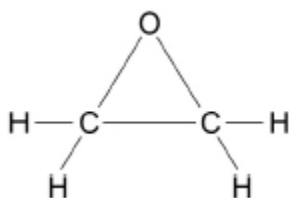
Epoxyethane is formed by the reaction of ethene with oxygen.

- (a) Identify the catalyst for this reaction.

\_\_\_\_\_

(1)

- (b) The displayed formula for epoxyethane is shown below.



Explain, by reference to its structure, why epoxyethane is a reactive molecule.

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(2)

- (c) In acidic conditions, epoxyethane reacts with water to form ethane-1,2-diol.

Outline a mechanism for this reaction.

(4)

- (d) One molecule of epoxyethane reacts with one molecule of an alcohol to form a compound with  $M_r$  of 90.0. Analysis showed that this compound contained 53.3% carbon, 11.1% hydrogen and 35.6% oxygen by mass.

Calculate the molecular formula of the compound formed.

Show your working.

Molecular formula of compound = \_\_\_\_\_

(3)

(Total 10 marks)

12

Ethanal is prepared by heating ethanol with potassium dichromate(VI) in the presence of sulfuric acid. **Figures 1** and **2** show two possible ways of heating this reaction mixture.

Figure 1

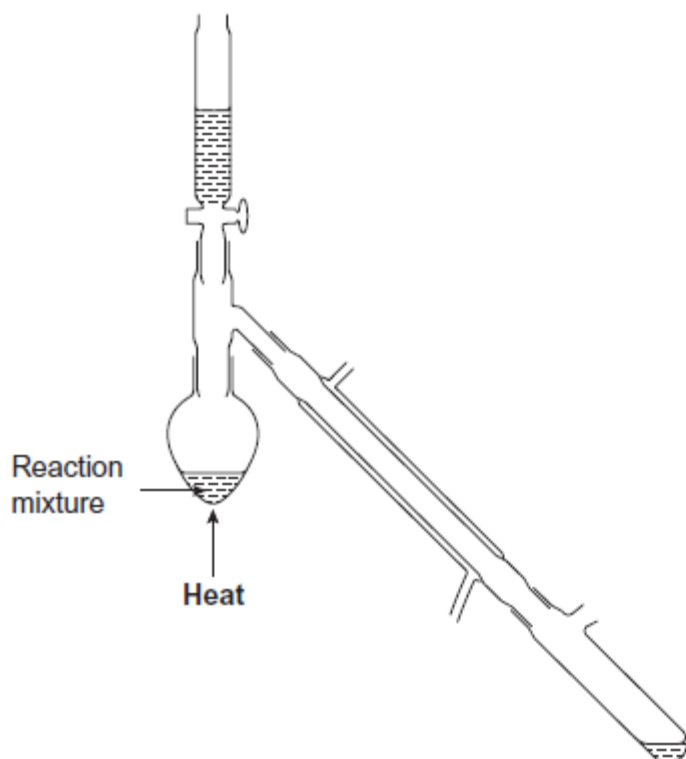
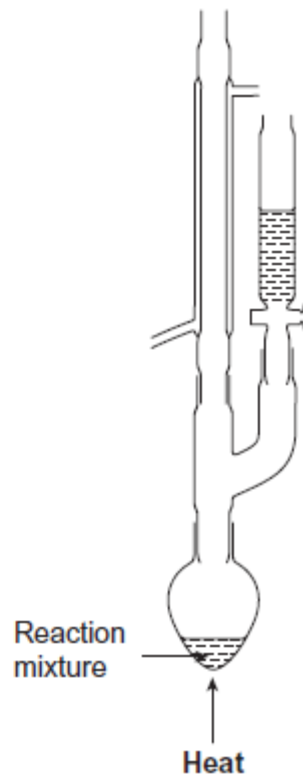


Figure 2



State which arrangement would **not** be suitable for the preparation of ethanal. Explain your answer.

Arrangement \_\_\_\_\_

Explanation \_\_\_\_\_

(Total 2 marks)

13

Ethanol can be oxidised slowly to ethanal. State how a sample of ethanol could be tested to confirm the presence of ethanal. State what you would observe.

Test \_\_\_\_\_

Observation \_\_\_\_\_

(Total 2 marks)

**14**

In each of the following questions, you should draw the structure of the compound in the space provided.

- (a) Draw the structure of the alkene that would form 1,2-dibromo-3-methylbutane when reacted with bromine.

(1)

- (b) Draw the structure of the alcohol with molecular formula  $C_4H_{10}O$  that is resistant to oxidation by acidified potassium dichromate(VI).

(1)

- (c) Draw the structure of the alkene that has a peak, due to its molecular ion, at  $m/z = 42$  in its mass spectrum.

(1)

- (d) Draw the structure of the organic product with  $M_r = 73$ , made from the reaction between 2-bromobutane and ammonia.

(1)

(Total 4 marks)

**15**

Compound **E** contains carbon, hydrogen and oxygen only.  
**E** contains 54.51% carbon and 9.15% hydrogen by mass.

- (a) Calculate the empirical formula of **E**.

Empirical formula \_\_\_\_\_

(3)

- (b) The mass spectrum of **E** showed a molecular ion peak with  $m/z = 88$

Use this information and your answer to question (a) to determine the molecular formula of **E**.

Molecular formula \_\_\_\_\_

(2)

- (c) The infrared spectrum of **E** shows

- a broad absorption in the region  $3230\text{--}3550\text{ cm}^{-1}$
- **no** absorption in the region  $1680\text{--}1750\text{ cm}^{-1}$

Use **Table A** in the Data Sheet to deduce what information the infrared spectrum provides about the structure of **E**.

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(2)

- (d) Outline a chemical test that would confirm that **E** contains a C=C double bond.

Test \_\_\_\_\_

Observation \_\_\_\_\_

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(2)

(Total 9 marks)

**16**

Which of these substances does **not** contribute to the greenhouse effect?

- A**      Unburned hydrocarbons.      ☐
- B**      Carbon dioxide.      ☐
- C**      Water vapour.      ☐
- D**      Nitrogen.      ☐

**(Total 1 mark)**

## Mark schemes

|   |                                                                                                                                                                                                                                                                                                                                                                                                                               |     |  |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--|
| 1 | C                                                                                                                                                                                                                                                                                                                                                                                                                             | [1] |  |
| 2 | <p>(a)</p> <ul style="list-style-type: none"> <li>• (Same) General formula/allow a named homologous series with its general formula</li> <li>• Chemically similar/same (chemical) reactions</li> <li>• Same functional group</li> <li>• <u>Trend</u> in physical properties/eg inc bp as <math>M_r</math> increases</li> <li>• (Molecules) increase by <math>\text{CH}_2/M_r = 14</math><br/><i>Any two points</i></li> </ul> | 2   |  |
|   | <p>(b) <u>Fractional</u> distillation/fractionation/chromatography<br/><i>Allow GLC</i></p>                                                                                                                                                                                                                                                                                                                                   | 1   |  |
|   | <p>(c) (Molecules/compounds/substances) with the same <u>molecular</u> formula/same number and type of atoms<br/><i>Allow alkanes with same molecular formula</i><br/><i>Allow same chemical formula in M1 = 0 but can allow M2</i></p>                                                                                                                                                                                       | 1   |  |
|   | <p>but different structural formula/different displayed formula/different arrangement of atoms/different structures<br/><i>Not different positions in space</i></p>                                                                                                                                                                                                                                                           | 1   |  |
|   | <p><u>2,4-dimethylhexane</u><br/><i>M2 dependent on M1</i></p>                                                                                                                                                                                                                                                                                                                                                                | 1   |  |
|   | <p><math>\text{C}_4\text{H}_9</math><br/><i>Ignore the absence of dash and/or commas</i></p>                                                                                                                                                                                                                                                                                                                                  | 1   |  |



- (d) less surface contact/less surface area/less polarisable molecule

1

so fewer/weaker/less Van der Waals'/vdw forces

*Allow more spherical or fewer points of contact*

*Not smaller molecule/not more compact molecule/not shorter chain*

*Allow converse arguments*

*Must be comparative answer ie not just few VDW forces*

*QoL*

*Assume 'it' refers to the branched isomer*

1

[9]

3 C

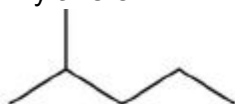
[1]

4 (a) Cyclohexane

*Accept branched cyclic alkanes eg methylcyclopentane or dimethylcyclobutane.*

1

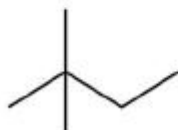
(b) Any **two** of:



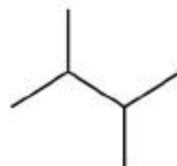
OR



OR



OR



2

Structural

1

(c)  $\text{C}_{22}\text{H}_{46} \rightarrow \text{C}_6\text{H}_{14} + 2 \text{C}_8\text{H}_{16}$

1

$P > 1000 \text{ kPa} / 10 \text{ atm}$

1

(d) Aluminosilicate / zeolite

1

(e)  $\text{Br}-\text{Br} \rightarrow 2\text{Br}^\bullet$

1

(f)  $\text{C}_3\text{H}_7^\bullet + \text{C}_3\text{H}_7^\bullet \rightarrow \text{C}_6\text{H}_{14}$

1

[9]

5

D

[1]

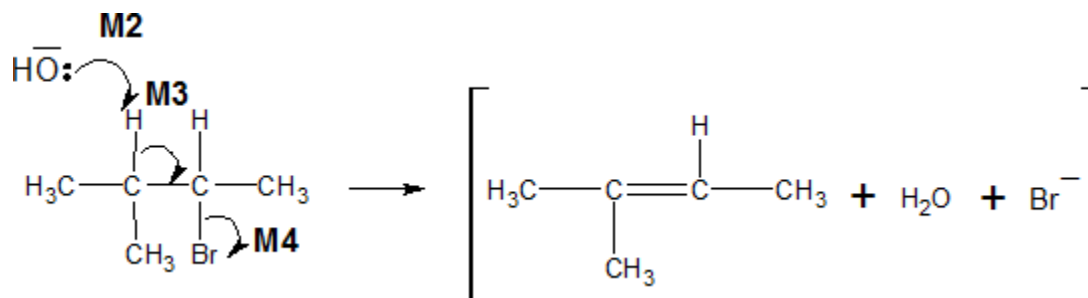
6

C

[1]

## (a) (i) M1 Elimination

M1 Credit "base elimination" but no other prefix.



M2 must show an arrow from the lone pair on the oxygen of a negatively charged hydroxide ion to a correct H atom

Ignore other partial charges

M3 must show an arrow from a correct C-H bond adjacent to the C-Br bond to a correct C-C bond. Only award if an arrow is shown attacking the H atom of a correct adjacent C-H bond in **M2**

Penalise **once only** in any part of the mechanism for a line and two dots to show a bond

M4 is independent provided it is from their original molecule, **BUT CE=0 for the mechanism (penalise M2, M3 and M4 only)** if nucleophilic substitution mechanism is shown

Maximum any 2 of 3 marks for the mechanism for wrong organic reactant or wrong organic product (if shown).

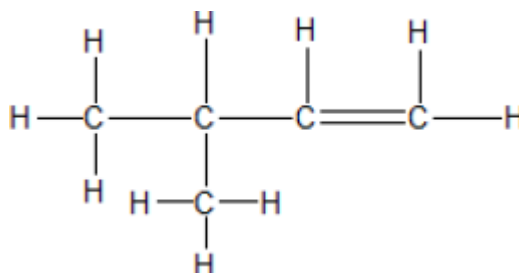
Credit the correct use of "sticks" for the molecule except for the C-H being attacked

Award full marks for an E1 mechanism in which **M4** is on the correct carbocation

Penalise **M4**, if an additional arrow is drawn from Br eg to K<sup>+</sup>

**NB These are double-headed arrows**

4

(ii) Displayed formula for 3-methylbut-1-ene

All bonds and atoms must be drawn out, but ignore bond angles

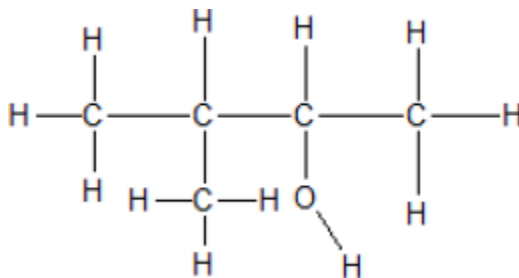
1

(iii) Position(al) (isomerism or isomer)

*Penalise any other words that are written in addition to these.*

1

(b) (i) Displayed formula for 3-methylbutan-2-ol



*All bonds and atoms must be drawn out, but ignore bond angles.*

1

(ii) Any **one** from

- Lower / decreased temperature **OR** cold
- Less concentrated (comparative) **OR** dilute KOH
- Water (as a solvent) / (aqueous conditions)  
*Ignore "pressure".*

1

(iii) Nucleophilic substitution

*Both words needed - credit phonetic spelling.*

1

(iv) (Strong / broad) absorption / peak in the range **3230 to 3550** cm<sup>-1</sup> or specified value in this range or marked correctly on spectrum

*Allow the words "dip" **OR** "spike" **OR** "trough" **OR** "low transmittance" as alternatives for absorption.*

1

[10]

8

C

[1]

9

B

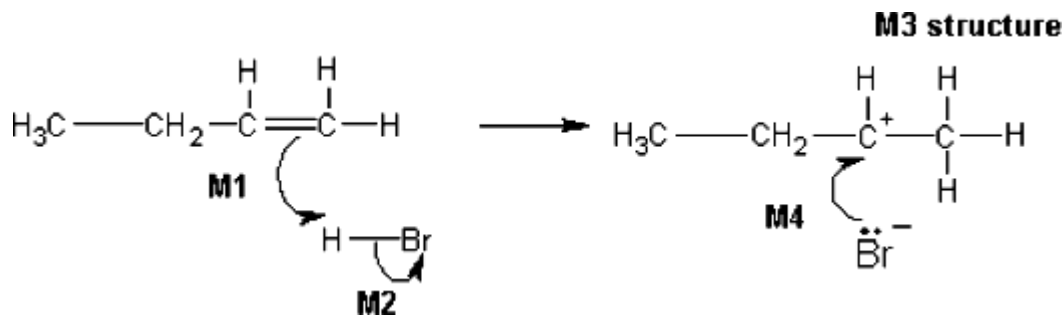
[1]

10

(a) Position(al) (isomerism)

1

(b)



**M1** must show an arrow from the double bond towards the H atom of the H–Br molecule

*M1 Ignore partial negative charge on the double bond.*

**M2** must show the breaking of the H–Br bond.

*M2 Penalise partial charges on H–Br bond if wrong way and penalise formal charges*

**M3** is for the structure of the secondary carbocation.

*Penalise M3 if there is a bond drawn to the positive charge*

**M4** must show an arrow from the lone pair of electrons on the negatively charged bromide ion towards the positively charged carbon atom of either a primary or secondary carbocation.

*Penalise once only in any part of the mechanism for a line and two dots to show a bond*

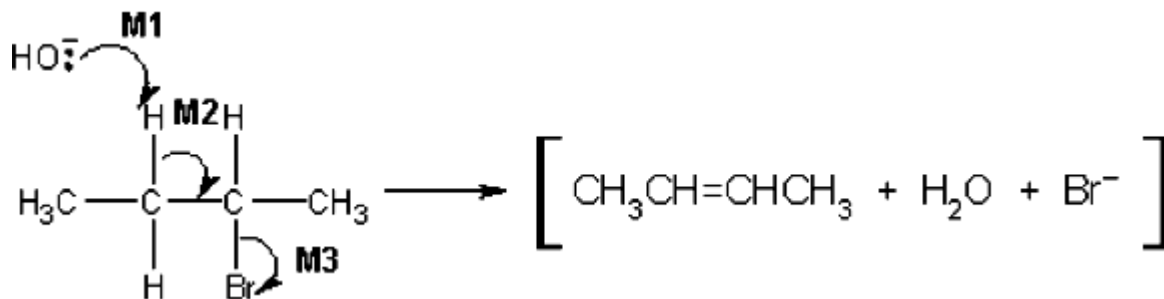
*Maximum any 3 of 4 marks for wrong reactant or primary carbocation.*

*If Br<sub>2</sub> is used, maximum 2 marks for their mechanism*

*Do not penalise the use of “sticks”*

**NB The arrows here are double-headed**

(c)



Penalise one mark from their total if half-headed arrows are used

**M1** must show an arrow from the lone pair on oxygen of a negatively charged hydroxide ion to a correct H atom

Penalise M1 if covalent KOH

**M2** must show an arrow from a  $\text{C-H}$  bond adjacent to the  $\text{C-Br}$  bond towards the appropriate  $\text{C-C}$  bond. Only award if an arrow is shown attacking the H atom of an adjacent  $\text{C-H}$  (in M1)

**M3** is independent provided it is from their original molecule.

Penalise M3 for formal charge on C of the  $\text{C-Br}$  or incorrect partial charges on  $\text{C-Br}$

Penalise M3 if an extra arrow is drawn from the Br of the  $\text{C-Br}$  bond to, for example,  $\text{K}^+$

Ignore other partial charges

Penalise once only in any part of the mechanism for a line and two dots to show a bond.

Maximum any 2 of 3 marks for wrong reactant or wrong product (if shown) or a mechanism that leads to but-1-ene

Accept the correct use of "sticks" for the molecule except for the  $\text{C-H}$  being attacked

Award full marks for an E1 mechanism in which M2 is on the correct carbocation.

**NB The arrows here are double-headed**

3

[8]

11

(a) Ag/silver

1

(b)  $\text{C-O-C}$  bond angle much smaller than  $109.5^\circ$

Allow "bond angle only  $60^\circ$ "

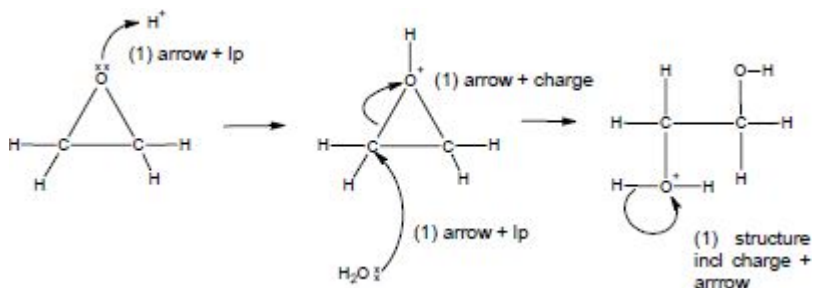
1

Bonding pairs repel each other

Allow "Ring strain"/lower  $\text{C-O}$  bond enthalpy

1

(c)



4

(d)

| C         | O         | H        |
|-----------|-----------|----------|
| 53.3/12.0 | 35.6/16.0 | 11.1/1.0 |
| 4.44      | 2.23      | 11.1     |
| 2         | 1         | 5        |

M1: divide by  $A_r$ 

M2: ratio 2:1:5

Empirical formula =  $\text{C}_2\text{OH}_5$ 90/45 = 2, so Molecular formula =  $\text{C}_4\text{O}_2\text{H}_{10}$ 

M3: molecular formula

3

[10]

12

Figure 2

1

Further oxidation will occur / ethanoic acid formed

Do not accept 'poor yield' without qualification

Can gain this mark if logic correct but has chosen wrong Figure

1

[2]

13

Add Tollens / Fehling's / Benedict's reagent / ir spectra

Accept any other chemically correct reagent and observation

1

Silver mirror / blue to red **OR** red precipitate (with ethanal) / peak at  $1700\text{ cm}^{-1}$  (in ethanal)

Must have correct test to access second mark

Accept 'silver'. Do not accept 'silver solution'

Give one mark for 'silver mirror test' and 'silver mirror'

Accept correct answer based on n.m.r. spectra

1

[2]

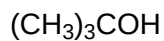
**14**

- (a) Structure for 3-methylbut-1-ene

*Any correct structural representation.**Credit "sticks" and require the double bond.*

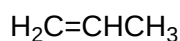
1

- (b) Structure for 2-methylpropan-2-ol

*Any correct structural representation.**Credit "sticks".*

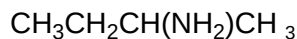
1

- (c) Structure for propene

*Any correct structural representation.**Credit "sticks" and require the double bond.*

1

- (d) Structure for 2-aminobutane

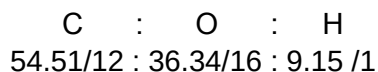
*Any correct structural representation.**Credit "sticks".*

1

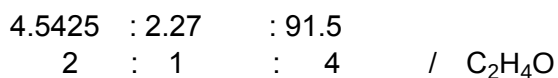
**[4]****15**

- (a) % O = 100 – 54.51 – 9.15 = 36.34

1



1



1

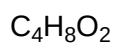
- (b) Molecular ion peak = 88 m/z

*Ecf from (a)*

Empirical mass = 44

1

88/44



1



(c) Broad 3300-3500 peak: -O-H / alcohol present

1

No 1680-1750 peak: **no** C=O / **no** carbonyl or acid group

1

(d) Test: addition of Br<sub>2</sub>

1

Observation: if C=C present, decolourisation.

1

**[9]**

**16**

D

**[1]**